

Question	Answer	Marks
6(a)	<u>sum of</u> e.m.f.(s) equal to <u>sum of</u> p.d.(s)	M1
	around a loop/around a closed circuit	A1
6(b)(i)	current in variable resistor = $(6.0/2.4) + (6.0/1.2)$ (= 7.5 A)	C1
	p.d. across variable resistor = $9.0 - 6.0$ (= 3.0 V)	C1
	$R = 3.0/7.5$ $= 0.40\Omega$	A1
	or	
	$\frac{1}{R_T} = \frac{1}{2.4} + \frac{1}{1.2}$ $R_T = 0.80 (\Omega)$	(C1)
	$\frac{3}{9} = \frac{R}{(R + 0.80)}$ or $\frac{3}{R} = \frac{6}{0.8}$	(C1)
6(b)(ii)	$P = V^2/R$ or $P = I^2R$ or $P = IV$	C1
	$P = 6.0^2/24$ or $2.5^2 \times 2.4$ or 6.0×2.5 $= 15W$	A1

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6(b)(iii)	1. $R = \frac{\rho L}{A}$	C1
	ratio = $(2.4/1.2) \times (3/1)$ = 6.0	A1
	2. $(I = nAvq)$ $I_X/I_Y = 2.5/5.0$ or $1.2/2.4$ or 0.5	C1
	ratio = $(2.5/5.0) \times (1/3)$ or $(1.2/2.4) \times (1/3)$ = 0.17	A1